

Group number: 08

Group member details: -

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A) Major factors for high-revenue orders

- Cylinders
- Fuel
- Transmission

Rationale: -

- Cylinders = 6 cylinders cars have higher share in high value orders
- Fuel = Gasoline cars have higher share in high value orders
- Transmission = Automatic and 8-speed automatic cars have higher share in high value orders

B) Recommendations: -

1. For improving product popularity and customer satisfaction: -

- i. Focusing on popular and high-revenue generating makes and models.
- ii. Manufacturing more SUV, Gasoline, all-wheel drive, Lesser Price because they have higher orders (demand) to reduce the waiting period.

2. Optimize pricing strategies for vehicles with a very high or low count of orders: -

- i. Vehicles with fuel as Diesel are more expensive as compared to others and have lesser demand, so reduce prices for these cars.

We have performed Data preprocessing, EDA and analysis using Python libraries and the same can be accessed after MongoDB results.

Starting MongoDB server: -

```
C:\Windows\System32\cmd.e X + v
Microsoft Windows [Version 10.0.22631.3810]
(c) Microsoft Corporation. All rights reserved.

C:\Program Files\MongoDB\Server\7.0\bin>mongod
{"t":{"sdate":"2024-07-06T13:54:58.741+05:30"},"s":"I", "c":"CONTROL", "id":23285, "ctx":"thread1","msg":"Automatically disabling TLS 1.0, to fo
rce-enable TLS 1.0 specify --sslDisabledProtocols 'none'"}
{"t":{"sdate":"2024-07-06T13:54:58.741+05:30"},"s":"I", "c":"NETWORK", "id":4915701, "ctx":"thread1","msg":"Initialized wire specification","attr
":{"spec":{"incomingExternalClient":{"minWireVersion":0,"maxWireVersion":21},"incomingInternalClient":{"minWireVersion":0,"maxWireVersion":21},"outgo
ing":{"minWireVersion":6,"maxWireVersion":21},"isInternalClient":true}}}}
{"t":{"sdate":"2024-07-06T13:54:58.743+05:30"},"s":"I", "c":"NETWORK", "id":4648602, "ctx":"thread1","msg":"Implicit TCP FastOpen in use."}
{"t":{"sdate":"2024-07-06T13:54:58.745+05:30"},"s":"I", "c":"REPL", "id":5123008, "ctx":"thread1","msg":"Successfully registered PrimaryOnlySer
vice", "attr":{"service":"TenantMigrationDonorService", "namespace":"config.tenantMigrationDonors"}}
{"t":{"sdate":"2024-07-06T13:54:58.745+05:30"},"s":"I", "c":"REPL", "id":5123008, "ctx":"thread1","msg":"Successfully registered PrimaryOnlySer
vice", "attr":{"service":"TenantMigrationRecipientService", "namespace":"config.tenantMigrationRecipients"}}
{"t":{"sdate":"2024-07-06T13:54:58.745+05:30"},"s":"I", "c":"CONTROL", "id":5945603, "ctx":"thread1","msg":"Multi threading initialized"}
{"t":{"sdate":"2024-07-06T13:54:58.745+05:30"},"s":"I", "c":"TENANT_M", "id":7091600, "ctx":"thread1","msg":"Starting TenantMigrationAccessBlockerR
egistry"}
{"t":{"sdate":"2024-07-06T13:54:58.746+05:30"},"s":"I", "c":"CONTROL", "id":4615611, "ctx":"initandlisten","msg":"MongoDB starting", "attr":{"pid":
11012, "port":27017, "dbPath":"C:/data/db/", "architecture":"64-bit", "host":"LAPTOP-IEG727EP"}}
{"t":{"sdate":"2024-07-06T13:54:58.746+05:30"},"s":"I", "c":"CONTROL", "id":23398, "ctx":"initandlisten","msg":"Target operating system minimum
version", "attr":{"targetMinOS":"Windows 7/Windows Server 2008 R2"}}
{"t":{"sdate":"2024-07-06T13:54:58.746+05:30"},"s":"I", "c":"CONTROL", "id":23403, "ctx":"initandlisten","msg":"Build Info", "attr":{"buildInfo":
{"version":"7.0.11", "gitVersion":"f451228f0df2b9dfe073f1521837f8ec5c208a8c", "modules":[], "allocator":"tcmalloc", "environment":{"distmod":"windows", "
distarch":"x86_64", "target_arch":"x86_64"}}}}
{"t":{"sdate":"2024-07-06T13:54:58.746+05:30"},"s":"I", "c":"CONTROL", "id":51765, "ctx":"initandlisten","msg":"Operating System", "attr":{"os":{"
name":"Microsoft Windows 10", "version":"10.0 (build 22631)}}}}
{"t":{"sdate":"2024-07-06T13:54:58.747+05:30"},"s":"I", "c":"CONTROL", "id":21951, "ctx":"initandlisten","msg":"Options set by command line", "at
tr":{"options":{}}}
{"t":{"sdate":"2024-07-06T13:54:58.750+05:30"},"s":"I", "c":"STORAGE", "id":22270, "ctx":"initandlisten","msg":"Storage engine to use detected b
y data files", "attr":{"dbpath":"C:/data/db/", "storageEngine":"wiredTiger"}}
{"t":{"sdate":"2024-07-06T13:54:58.751+05:30"},"s":"I", "c":"STORAGE", "id":22315, "ctx":"initandlisten","msg":"Opening WiredTiger", "attr":{"con
fig":{"create,cache_size=3495M,session_max=33000,eviction=(threads_min=4,threads_max=4),config_base=false,statistics=(fast),log=(enabled=true,remove=
true,path=journal,compressor=snappy),builtin_extension_config=(zstd=(compression_level=6)),file_manager=(close_idle_time=600,close_scan_interval=10,
close_handle_minimum=2000),statistics_log=(wait=0),json_output=(error,message),verbose=[recovery_progress:1,checkpoint_progress:1,compact_progress:1
,backup:0,checkpoint:0,compact:0,evict:0,history_store:0,recovery:0,rts:0,salvage:0,tiered:0,timestamp:0,transaction:0,verify:0,log:0],"}}}}

```

Starting MongoDB shell: -

```
mon. mongosh mongodb://127.0.0. X + v
Microsoft Windows [Version 10.0.22631.3810]
(c) Microsoft Corporation. All rights reserved.

C:\Program Files\MongoDB\Server\7.0\bin>mongosh
Current Mongosh Log ID: 6688ff6f5150d6d46ccc8987
Connecting to: mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2
.2.10
Using MongoDB: 7.0.11
Using Mongosh: 2.2.10

For mongosh info see: https://docs.mongodb.com/mongosh-shell/

-----
The server generated these startup warnings when booting
2024-06-29T14:58:45.056+05:30: Access control is not enabled for the database. Read and write access to data and conf
iguration is unrestricted
-----

test>
```

Fetching list of databases in MongoDB

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.2.10
Microsoft Windows [Version 10.0.22631.3810]
(c) Microsoft Corporation. All rights reserved.

C:\Program Files\MongoDB\Server\7.0\bin>mongosh
Current Mongosh Log ID: 6688ff6f5150d6d46ccc8987
Connecting to:  mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.2.10
Using MongoDB:  7.0.11
Using Mongosh:  2.2.10

For mongosh info see: https://docs.mongodb.com/mongodb-shell/

-----
  The server generated these startup warnings when booting
  2024-06-29T14:58:45.056+05:30: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
-----

test> show dbs
BDS_assignment_1  648.00 KiB
admin             40.00 KiB
config            72.00 KiB
local             72.00 KiB
vehicle_dataset   644.00 KiB
test>
```

Accessing the database: BDS_assignment_1

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.2.10
Microsoft Windows [Version 10.0.22631.3810]
(c) Microsoft Corporation. All rights reserved.

C:\Program Files\MongoDB\Server\7.0\bin>mongosh
Current Mongosh Log ID: 6688ff6f5150d6d46ccc8987
Connecting to:  mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.2.10
Using MongoDB:  7.0.11
Using Mongosh:  2.2.10

For mongosh info see: https://docs.mongodb.com/mongodb-shell/

-----
  The server generated these startup warnings when booting
  2024-06-29T14:58:45.056+05:30: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
-----

test> show dbs
BDS_assignment_1  648.00 KiB
admin             40.00 KiB
config            72.00 KiB
local             72.00 KiB
vehicle_dataset   644.00 KiB
test> use BDS_assignment_1
switched to db BDS_assignment_1
BDS_assignment_1>
```

Q1. Calculate the number of vehicles which are Jeeps.

```
mongosh mongodb://127.0.0.1 x + v
BDS_assignment_1> db.vehicle.countDocuments({'make':'Jeep'})
192
BDS_assignment_1> |
```

Q2. Write a query to find no of vehicles which are SUV's

```
mongosh mongodb://127.0.0.1 x + v
BDS_assignment_1> db.vehicle.countDocuments({'body':'SUV'})
690
BDS_assignment_1>
```

Q3. Find the number of vehicles which have less than four doors in the entire dataset.

```
mongosh mongodb://127.0.0.1 x + v
BDS_assignment_1> db.vehicle.countDocuments({'doors':{'$lt':4}})
47
BDS_assignment_1>
```

Q4. Write a query to find the number of vehicles which have mileage less than 20,000.

```
mongosh mongodb://127.0.0.1 x + v
BDS_assignment_1> db.vehicle.countDocuments({'mileage':{'$lt':20000}})
978
BDS_assignment_1>
```

Q5. Retrieve the details of make and mileage of the first vehicle.

```
mongosh mongodb://127.0.0.1 x + v
BDS_assignment_1> db.vehicle.findOne({},{'make':1, 'mileage':1})
{
  _id: ObjectId('668abdd905bcff8cbfb7617b'),
  make: 'Jeep',
  mileage: 10
}
BDS_assignment_1>
```

Q6. Find the number of vehicles bought in the year 2023.

```
mongosh mongodb://127.0.0.1 x + v
BDS_assignment_1> db.vehicle.countDocuments({'year':{'$eq':2023}})
82
BDS_assignment_1>
```

Q7. Find the maximum no. of doors that a car can have.

```
mongosh mongodb://127.0.0.1:27010/
BDS_assignment_1> db.vehicle.aggregate([ { $group: { _id:null, MaximumValue: { $max: "$doors" } } } ])
[ { _id: null, MaximumValue: 5 } ]
BDS_assignment_1>
```

Q8. Calculate the average price of SUV cars.

```
mongosh mongodb://127.0.0.1:27010/
BDS_assignment_1> db.vehicle.aggregate([{$match: {body: "SUV"}},{ $group: { _id: null, average_price: { $avg: "$price" } }}
])
[ { _id: null, average_price: 47854.11304347826 } ]
BDS_assignment_1>
```

Q9. Identify the no of vehicles with price greater than 40,000 less than 70,000.

```
mongosh mongodb://127.0.0.1:27010/
BDS_assignment_1> db.vehicle.countDocuments({'price':{'$lt:70000,$gt:40000'}})
545
BDS_assignment_1>
```

Q10. Calculate the no of vehicles with fuel type only hybrid (not PHEV Hybrid)

```
mongosh mongodb://127.0.0.1:27010/
BDS_assignment_1> db.vehicle.countDocuments({'fuel':'Hybrid'})
136
BDS_assignment_1>
```

Q11. Find the no of fuel types.

```
mongosh mongodb://127.0.0.1:27010/
BDS_assignment_1> db.vehicle.distinct("fuel").length
7
BDS_assignment_1>
```

Q12. Retrieve the details (make,model,type,year,price,mileage) of jeeps which have white as exterior_colour, Global black as interior colour.

```
mongosh mongodb://127.0.0.1:27027/
BDS_assignment_1> db.vehicle.find({"exterior_color":"White","interior_color":"Global Black"},{"make":1,"model":1,"type":1,"year":1,"price":1,"mileage":1})
[
  {
    _id: ObjectId('668abdd905bcff8cbfb7617b'),
    make: 'Jeep',
    model: 'Wagoneer',
    type: 'New',
    year: 2024,
    price: 74600,
    mileage: 10
  }
]
BDS_assignment_1> |
```

Loading libraries

```
In [1]: 1 import pandas as pd
        2 import numpy as np
        3 import matplotlib.pyplot as plt
        4 import seaborn as sns
```

Loading the dataset

```
In [2]: 1 vehicle_df=pd.read_csv("vehicles_dataset.csv")
        2 vehicle_orig_df=vehicle_df.copy(deep=True)
```

Checking initial rows and final rows

```
In [3]: 1 vehicle_df.head()
```

Out[3]:

	name	description	make	model	type	year	price	engine	cylinders	fuel
0	2024 Jeep Wagoneer Series II	Heated Leather Seats, Nav System, Moonroof, Th...	Jeep	Wagoneer	New	2024	74600.0	24V GDI DOHC Twin Turbo	6.0	Gasoline
1	2024 Jeep Grand Cherokee Laredo	Al West is committed to offering every custome...	Jeep	Grand Cherokee	New	2024	50170.0	OHV	6.0	Gasoline
2	2024 GMC Yukon XL Denali	unknown	GMC	Yukon XL	New	2024	96410.0	6.2L V-8 gasoline direct injection, variable v...	8.0	Gasoline
3	2023 Dodge Durango Pursuit	White Knuckle Clearcoat 2023 Dodge Durango Pur...	Dodge	Durango	New	2023	46835.0	16V MPFI OHV	8.0	Gasoline
4	2024 RAM 3500 Laramie	2024 Ram 3500 Laramie Billet Silver Metallic C...	RAM	3500	New	2024	81663.0	24V DDI OHV Turbo Diesel	6.0	Diese

In [4]:

```
1 vehicle_df.tail()
```

Out[4]:

	name	description	make	model	type	year	price	engine	cylinders
997	2024 Mercedes-Benz Sprinter 2500 Standard Roof	2024 Mercedes-Benz Sprinter 2500 Cargo 144 WB ...	Mercedes-Benz	Sprinter 2500	New	2024	59037.0	16V DDI DOHC Turbo Diesel	4.0
998	2024 Dodge Hornet R/T Plus Eawd	Dealer Comments +++ Price Ends 5/31/2024 +++ A...	Dodge	Hornet	New	2024	49720.0	4 gasoline direct injection, DOHC, Multiair va...	4.0 (
999	2024 Jeep Wagoneer Base	The ALL New Friendship CDJR of Johnson City of...	Jeep	Wagoneer	New	2024	69085.0	24V GDI DOHC Twin Turbo	6.0 (
1000	2024 Nissan Murano SV Intelligent AWD	CVT with Xtronic, AWD.At Todd Nissan Of Torrin...	Nissan	Murano	New	2024	43495.0	6 DOHC, variable valve control, regular unlead...	6.0 (
1001	2024 Chevrolet Silverado 2500 WT	01u 2024 Chevrolet Silverado 2500HD Work Truck...	Chevrolet	Silverado 2500	New	2024	48995.0	8 gasoline direct injection, variable valve co...	8.0 (

Data Cleaning and Preprocessing

```
In [5]: 1 #Checking the dataset information
        2 vehicle_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1002 entries, 0 to 1001
Data columns (total 18 columns):
#   Column                Non-Null Count  Dtype
---  -
0   name                   1002 non-null   object
1   description             1002 non-null   object
2   make                   1002 non-null   object
3   model                  1002 non-null   object
4   type                   1002 non-null   object
5   year                   1002 non-null   int64
6   price                  1002 non-null   float64
7   engine                 1002 non-null   object
8   cylinders              1002 non-null   float64
9   fuel                  1002 non-null   object
10  mileage                1002 non-null   float64
11  transmission           1002 non-null   object
12  trim                   1002 non-null   object
13  body                   1002 non-null   object
14  doors                  1002 non-null   float64
15  exterior_color         1002 non-null   object
16  interior_color         1002 non-null   object
17  drivetrain             1002 non-null   object
dtypes: float64(4), int64(1), object(13)
memory usage: 141.0+ KB
```

```
In [6]: 1 #Checking for number of null values in each field
        2 vehicle_df.isnull().sum()
```

```
Out[6]: name                0
        description         0
        make                0
        model               0
        type                0
        year                0
        price               0
        engine              0
        cylinders            0
        fuel                0
        mileage             0
        transmission        0
        trim                0
        body                0
        doors               0
        exterior_color      0
        interior_color      0
        drivetrain          0
dtype: int64
```

```
In [7]: 1 #Checking for number of na values in each field
        2 vehicle_df.isna().sum()
```

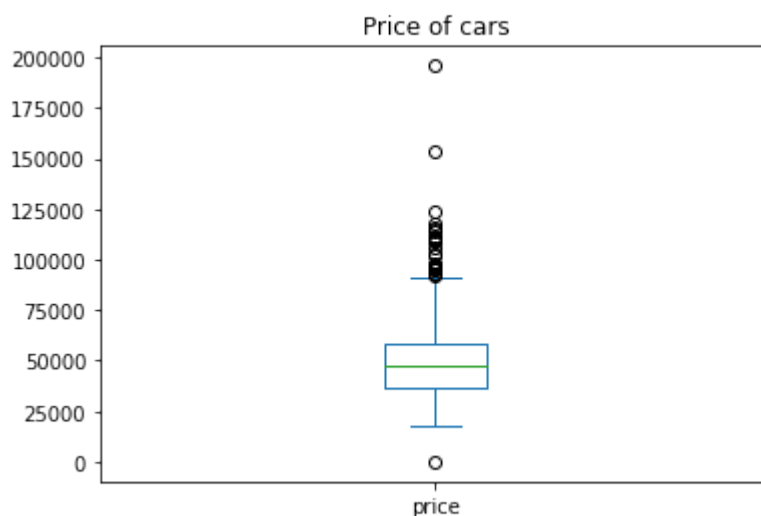
```
Out[7]: name          0
        description    0
        make           0
        model          0
        type           0
        year           0
        price          0
        engine         0
        cylinders      0
        fuel           0
        mileage        0
        transmission   0
        trim           0
        body           0
        doors          0
        exterior_color 0
        interior_color 0
        drivetrain     0
        dtype: int64
```

The list of features having null values: -

1. description
2. price
3. engine
4. cylinders
5. fuel
6. mileage
7. transmission
8. trim
9. body
10. exterior_color
11. interior_color

```
In [8]: 1 #Plotting distribution of price
        2 vehicle_df['price'].plot(kind='box', title='Price of cars')
```

```
Out[8]: <AxesSubplot: title={'center': 'Price of cars'}>
```



- We observed the distribution of values under the field: price is highly skewed.
- Thus, we replace all null values in the field with median value.

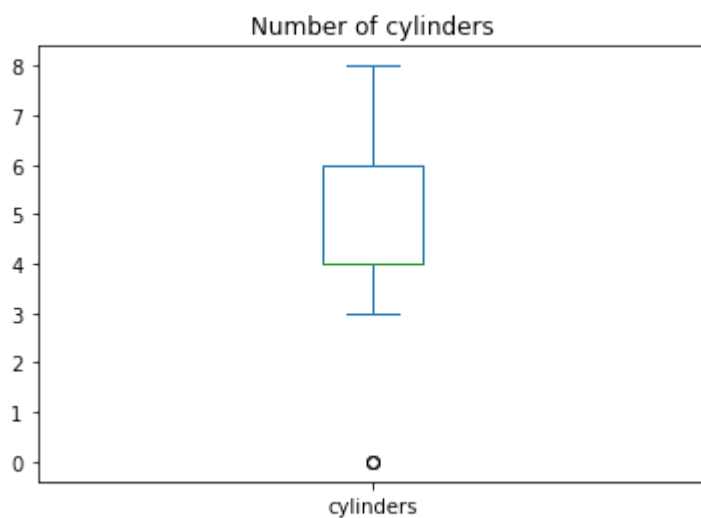
```
In [9]: 1 vehicle_df['price'].fillna(vehicle_df['price'].median(), inplace=True)
```

```
In [10]: 1 vehicle_df.isna().sum()
```

```
Out[10]: name                0
         description         0
         make                0
         model               0
         type                0
         year                0
         price               0
         engine              0
         cylinders           0
         fuel                0
         mileage             0
         transmission        0
         trim                0
         body                0
         doors               0
         exterior_color       0
         interior_color       0
         drivetrain           0
         dtype: int64
```

```
In [11]: 1 #Plotting distribution of cylinders
         2 vehicle_df['cylinders'].plot(kind='box', title='Number of cylinders')
```

```
Out[11]: <AxesSubplot: title={'center': 'Number of cylinders'}>
```



- We observed the median value is very close to lower quartile.
- Thus, we replace the missing values with median.

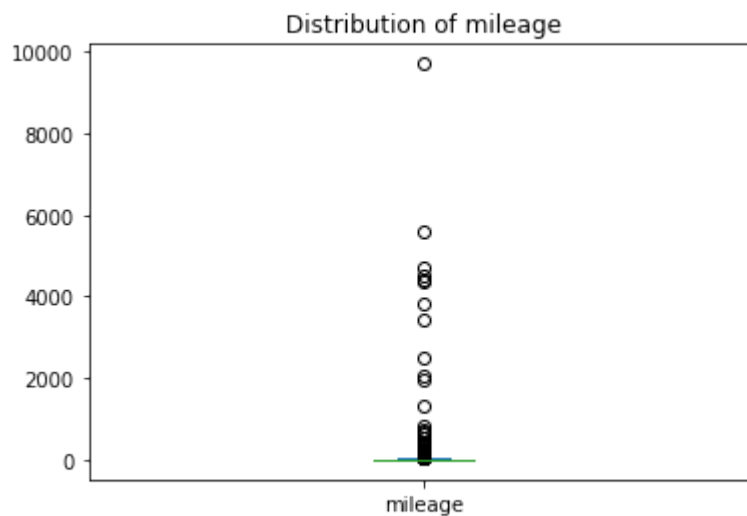
```
In [12]: 1 vehicle_df['cylinders'].fillna(vehicle_df['cylinders'].median(), inplace=True)
```

```
In [13]: 1 vehicle_df.isna().sum()
```

```
Out[13]: name          0
         description    0
         make          0
         model         0
         type          0
         year          0
         price         0
         engine        0
         cylinders     0
         fuel          0
         mileage       0
         transmission  0
         trim          0
         body          0
         doors         0
         exterior_color 0
         interior_color 0
         drivetrain    0
         dtype: int64
```

```
In [14]: 1 #Plotting distribution of mileage
         2 vehicle_df['mileage'].plot(kind='box', title='Distribution of mileage')
```

```
Out[14]: <AxesSubplot: title={'center': 'Distribution of mileage'}>
```



- We observed the mileage is highly skewed.

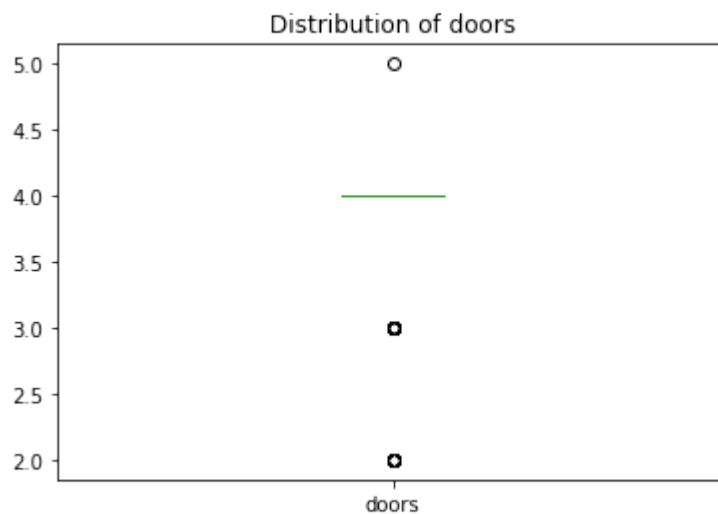
```
In [15]: 1 vehicle_df['mileage'].fillna(vehicle_df['mileage'].median(), inplace=True)
```

```
In [16]: 1 vehicle_df.isna().sum()
```

```
Out[16]: name                0
         description         0
         make                0
         model               0
         type                0
         year                0
         price               0
         engine              0
         cylinders           0
         fuel                0
         mileage             0
         transmission        0
         trim                0
         body                0
         doors               0
         exterior_color      0
         interior_color      0
         drivetrain          0
         dtype: int64
```

```
In [17]: 1 #Plotting distribution of doors
        2 vehicle_df['doors'].plot(kind='box', title='Distribution of doors')
```

```
Out[17]: <AxesSubplot: title={'center': 'Distribution of doors'}>
```



- We observed median with very close to lower and upper quartile values.

```
In [18]: 1 vehicle_df['doors'].fillna(vehicle_df['doors'].median(), inplace=True)
```

```
In [19]: 1 vehicle_df.isna().sum()
```

```
Out[19]: name                0
description                0
make                      0
model                    0
type                     0
year                     0
price                    0
engine                   0
cylinders                0
fuel                    0
mileage                  0
transmission             0
trim                     0
body                     0
doors                    0
exterior_color            0
interior_color            0
drivetrain               0
dtype: int64
```

```
In [20]: 1 vehicle_description=vehicle_df.groupby('description')['description'].count()
2         vehicle_description_df=pd.DataFrame({'description':vehicle_description.index, 'count':vehicle_description.values})
3         vehicle_description_df
```

```
Out[20]:
```

	description	count
0	!!! CALL US AT (812) 477-6201 !!!2024 Honda CR...	2
1	!!! CALL US AT (812) 477-6201 !!!2024 Honda CR...	1
2	!!! CALL US AT (812) 477-6201 !!!2025 Honda CR...	1
3	\$1,042 off MSRP!Race Red 2024 Ford F-350SD XL ...	1
4	\$1,300 off MSRP! Apple CarPlay/Android Auto, H...	1
...	...	...
757	You won't want to miss this excellent value! T...	1
758	Your satisfaction is our business!? Check out ...	1
759	must finance with vci using standard ratesCar ...	1
760	unknown	56
761	~BACK UP CAMERA~, AWD, ~BLUETOOTH, HANDS-FREE ...	1

762 rows × 2 columns

```
In [21]: 1 #We replace all empty cells with "Unknown"
2         vehicle_df["description"].fillna("unknown", inplace = True)
```

```
In [22]: 1 vehicle_df.isna().sum()
```

```
Out[22]: name                0
description            0
make                  0
model                 0
type                  0
year                  0
price                 0
engine                0
cylinders              0
fuel                  0
mileage               0
transmission          0
trim                  0
body                  0
doors                 0
exterior_color        0
interior_color        0
drivetrain            0
dtype: int64
```

```
In [23]: 1 vehicle_engine=vehicle_df.groupby('engine')['engine'].count()
2 vehicle_engine_df=pd.DataFrame({'engine':vehicle_engine.index, 'count':
3 vehicle_engine_df
```

```
Out[23]:
```

	engine	count
0	.6L I-4 gasoline direct injection, DOHC, D-CVV...	2
1	1.3L I-3 gasoline direct injection, DOHC, vari...	1
2	12V GDI DOHC Turbo	6
3	12V GDI OHV	1
4	12V PDI DOHC Turbo	6
...	...	...
96	ream 1.6L I-4 gasoline direct injection, DOHC,...	5
97	ream 2.5L I-4 port/direct injection, DOHC, CVV...	1
98	t 1.5L I-3 port/direct injection, DOHC, Ti-VCT...	6
99	t 2L I-4 gasoline direct injection, DOHC, vari...	4
100	unknown	2

101 rows × 2 columns

```
In [24]: 1 vehicle_df[vehicle_df['engine'].isnull()]
```

```
Out[24]:
```

	name	description	make	model	type	year	price	engine	cylinders	fuel	mileage	transn
0												

```
In [25]: 1 #We observed among the records with field: engine having no value, the
2 #Thus, we replace empty value in the field with: Unknown
3 vehicle_df["engine"].fillna("unknown", inplace = True)
```

```
In [26]: 1 vehicle_df.isna().sum()
```

```
Out[26]: name          0
description  0
make        0
model       0
type        0
year        0
price       0
engine      0
cylinders   0
fuel        0
mileage     0
transmission 0
trim        0
body        0
doors       0
exterior_color 0
interior_color 0
drivetrain   0
dtype: int64
```

```
In [27]: 1 vehicle_fuel=vehicle_df.groupby('fuel')['fuel'].count()
2 vehicle_fuel_df=pd.DataFrame({'fuel':vehicle_fuel.index, 'count':vehicle_fuel['count']}
3 vehicle_fuel_df
```

```
Out[27]:
```

	fuel	count
0	Diesel	73
1	Diesel (B20 capable)	1
2	E85 Flex Fuel	5
3	Electric	99
4	Gasoline	671
5	Hybrid	137
6	PHEV Hybrid Fuel	16

```
In [28]: 1 vehicle_df[vehicle_df['fuel'].isnull()]
```

```
Out[28]:
```

	name	description	make	model	type	year	price	engine	cylinders	fuel	mileage	transmission
--	------	-------------	------	-------	------	------	-------	--------	-----------	------	---------	--------------

- Out of 7 vehicles, 4 have make=Honda.
- All 4 vehicles with make=Honda, have model=Prologue, type=New, year=2024, cylinders=4.0, body=SUV, doors=4.0
- The other three vehicles have make=Subaru, Chevrolet and Jaguar.

```
In [29]: 1 vehicle_df.loc[(vehicle_df['make']=='Honda') & (vehicle_df['model']=='Prologue') &
2 (vehicle_df['type']=='New') & (vehicle_df['year']==2024) &
3 (vehicle_df['cylinders']==4.0) & (vehicle_df['body']=='SUV') &
4 (vehicle_df['doors']==4.0)]
```

```
Out[29]:
```

	name	description	make	model	type	year	price	engine	cylinders	fuel
219	2024 Honda Prologue TOURING	Shop Your Way Hoffman Honda wants you to shop ...	Honda	Prologue	New	2024	55800.0	c	4.0	Gasoline
315	2024 Honda Prologue TOURING	Shop Your Way Hoffman Honda wants you to shop ...	Honda	Prologue	New	2024	56550.0	c	4.0	Gasoline
489	2024 Honda Prologue TOURING	Shop Your Way Hoffman Honda wants you to shop ...	Honda	Prologue	New	2024	55800.0	c	4.0	Gasoline
490	2024 Honda Prologue TOURING	Shop Your Way Hoffman Honda wants you to shop ...	Honda	Prologue	New	2024	55800.0	c	4.0	Gasoline

Here we observed there no additional records with the above condition from which we can deduce the value for field: fuel.

In [30]:

1 vehicle_df.loc[(vehicle_df['make']=='Honda')]

Out[30]:

	name	description	make	model	type	year	price	engine	cylinder
109	2024 Honda CR-V Hybrid Sport Touring	!!! CALL US AT (812) 477-6201 !!!2024 Honda CR...	Honda	CR-V Hybrid	New	2024	42005.0	16V GDI DOHC Hybrid	4
219	2024 Honda Prologue TOURING	Shop Your Way Hoffman Honda wants you to shop ...	Honda	Prologue	New	2024	55800.0	c	4
257	2024 Honda CR-V Hybrid Sport-L	Still Night Pearl 2024 Honda CR- V Hybrid Sport...	Honda	CR-V	New	2024	38855.0	gasoline direct injection, DOHC, i-VTEC variab...	4

We observed most vehicles with make=Honda have fuel=Gasoline.

In [31]:

```
1 vehicle_df.loc[(vehicle_df['make']=='Subaru')]
```

Out[31]:

	name	description	make	model	type	year	price	engine	cylinders	fuel
47	2024 Subaru Outback Wilderness	Heated Seats, Alloy Wheels, Turbo Charged, All...	Subaru	Outback	New	2024	44354.0	16V GDI DOHC Turbo	4.0	Gasoline
128	2024 Subaru Solterra Premium	2024 Subaru Solterra Galactic Black Fox Subaru...	Subaru	Solterra	New	2024	39934.0	c	4.0	Gasoline
279	2024 Subaru Legacy Premium	4-Wheel Disc Brakes, 6 Speakers, ABS brakes, A...	Subaru	Legacy	New	2024	31042.0	16V GDI DOHC	4.0	Gasoline
476	2024 Subaru Outback Wilderness	Heated Seats, Alloy Wheels, Turbo, All Wheel D...	Subaru	Outback	New	2024	44354.0	16V GDI DOHC Turbo	4.0	Gasoline
801	2024 Subaru Outback Premium	Heated Seats, Alloy Wheels, All Wheel Drive, B...	Subaru	Outback	New	2024	32490.0	16V GDI DOHC	4.0	Gasoline
913	2024 Subaru Impreza Sport	2024 Subaru Impreza Pure Red Sport 27/34 City/...	Subaru	Impreza	New	2024	29068.0	16V GDI DOHC	4.0	Gasoline
922	2024 Subaru Outback Limited	Heated Leather Seats, Alloy Wheels, All Wheel ...	Subaru	Outback	New	2024	39822.0	16V GDI DOHC	4.0	Gasoline
946	2024 Subaru Impreza RS	2024 Subaru Impreza Crystal Black Silica 2.5RS...	Subaru	Impreza	New	2024	32098.0	16V GDI DOHC	4.0	Gasoline



In [32]: 1 vehicle_df.loc[(vehicle_df['make']=='Chevrolet')]

Out[32]:

	name	description	make	model	type	year	price	
14	2024 Chevrolet Blazer EV 2LT	Sterling Gray Metallic 2024 Chevrolet Blazer E...	Chevrolet	Blazer EV	New	2024	51695.0	
28	2024 Chevrolet Blazer EV 2LT	Radiant Red Tintcoat 2024 Chevrolet Blazer EV ...	Chevrolet	Blazer EV	New	2024	52190.0	
36	2024 Chevrolet Silverado 1500 RST	Summit White 2024 Chevrolet Silverado 1500 RST...	Chevrolet	Silverado 1500	New	2024	58610.0	16V GDI
38	2024 Chevrolet Blazer 3LT	Heated Leather Seats, NAV, Sunroof, Onboard Co...	Chevrolet	Blazer	New	2024	43727.0	24V GDI
	2024	IN STOCK!						

In [33]: 1 vehicle_df.loc[(vehicle_df['make']=='Jaguar')]

Out[33]:

	name	description	make	model	type	year	price	engine	cylinders
726	2024 Jaguar I-PACE EV400 R-Dynamic HSE	Contact our Sales Team at(708)342-0400 for mor...	Jaguar	I-PACE	New	2024	77053.0	d>\n\n \n <dt>VIN</dt>\n SADHM2S12R1...	4.0 G

Based on above observations, we replace all empty values in the field: fuel with Gasoline.

In [34]: 1 vehicle_df["fuel"].fillna("Gasoline", inplace = True)

```
In [35]: 1 vehicle_df.isna().sum()
```

```
Out[35]: name                0  
description                0  
make                      0  
model                     0  
type                      0  
year                      0  
price                     0  
engine                    0  
cylinders                 0  
fuel                      0  
mileage                   0  
transmission              0  
trim                      0  
body                      0  
doors                     0  
exterior_color            0  
interior_color            0  
drivetrain                0  
dtype: int64
```

```
In [36]: 1 vehicle_transmission=vehicle_df.groupby('transmission')['transmission']  
2 vehicle_transmission_df=pd.DataFrame({'transmission':vehicle_transmission  
3 vehicle_transmission_df
```

Out[36]:

	transmission	count
0	(CVT) CONT VAR.	1
1	1-Speed Automatic	69
2	1-Speed CVT with Overdrive	1
3	10-Speed Automatic	58
4	10-Speed Automatic with Overdrive	1
5	10-Speed Shiftable Automatic	1
6	6-SPEED AUTOMATIC	1
7	6-Spd Aisin F21-250 PHEV Auto Trans	7
8	6-Speed A/T	1
9	6-Speed Automatic	98
10	6-Speed Automatic Electronic with Overdrive	4
11	6-Speed DCT Automatic	1
12	62 kWh battery	1
13	7-Speed Automatic S tronic	4
14	7-Speed Automatic with Auto-Shift	4
15	7-Speed DSG Automatic with Tiptronic	4
16	7-Speed DSGA? Automatic w/ 4MO	11
17	8 Speed Dual Clutch	1
18	8-Speed A/T	4
19	8-Speed Automatic	325
20	8-Speed Automatic Sport	1
21	8-Speed Automatic with Auto-Shift	17
22	8-Speed Automatic with Tiptronic	10
23	8-Speed Shiftable Automatic	1
24	8-speed automatic	5
25	9 Spd Automatic	2
26	9-Speed 948TE Automatic	4
27	9-Speed A/T	1
28	9-Speed Automatic	62
29	9-speed automatic	1
30	A/T	1
31	Aisin 6-Speed Automatic	2
32	Automatic	238
33	Automatic CVT	45
34	CVT	9
35	CVT with Xtronic	2
36	Variable	3
37	automatic w/paddle shifters	1

```
In [37]: 1 vehicle_df[vehicle_df['transmission'].isnull()]
```

```
Out[37]:
```

	name	description	make	model	type	year	price	engine	cylinders	fuel	mileage	trans
--	------	-------------	------	-------	------	------	-------	--------	-----------	------	---------	-------

- The field: transmission has two null records.
- Record 1: make=Mercedes-Benz, year=2024, engine=c, fuel=Electric, body=Sedan, drivetrain=All-wheel Drive
- Record 2: make=Ford, year=2024, engine=24V PDI DOHC Flexible Fuel, fuel=E85 Flex Fuel, body=Cargo Van, drivetrain=Rear-wheel Drive

```
In [38]: 1 vehicle_df.loc[(vehicle_df['make']=='Mercedes-Benz') &  
2                 (vehicle_df['year']==2024) &  
3                 (vehicle_df['engine']=='c') &  
4                 (vehicle_df['fuel']=='Electric') &  
5                 (vehicle_df['body']=='Sedan') &  
6                 (vehicle_df['drivetrain']=='All-wheel Drive')]
```

```
Out[38]:
```

	name	description	make	model	type	year	price	engine	cylinders	fu
142	2024 Mercedes-Benz EQS 450 Base 4MATIC	Fletcher Jones Motorcars does not reserve or h...	Mercedes-Benz	EQS 450	New	2024	47165.0	c	4.0	Elect
253	2024 Mercedes-Benz EQS 450 Base 4MATIC	Panoramic Roof,Leather Seats,Rear Lip Spoiler,...	Mercedes-Benz	EQS 450	New	2024	110395.0	c	4.0	Elect
328	2024 Mercedes-Benz EQS 450 Base 4MATIC	Fletcher Jones Motorcars does not reserve or h...	Mercedes-Benz	EQS 450	New	2024	47165.0	c	4.0	Elect
372	2024 Mercedes-Benz EQS 450 Base 4MATIC	Fletcher Jones Motorcars does not reserve or h...	Mercedes-Benz	EQS 450	New	2024	47165.0	c	4.0	Elect
484	2024 Mercedes-Benz EQS 450 Base 4MATIC	This Mercedes-Benz EQS boasts a Electric engin...	Mercedes-Benz	EQS 450	New	2024	117985.0	c	4.0	Elect
725	2024 Mercedes-Benz EQS 450 Base 4MATIC	NAV, Sunroof, Heated Leather Seats, Alloy Whee...	Mercedes-Benz	EQS 450	New	2024	111245.0	c	4.0	Elect

- Here we observed all the records have transmission=Automatic.
- Thus, we will replace the empty value as Automatic.

```
In [39]: 1 vehicle_df.loc[(vehicle_df['make']=='Ford') &
2             (vehicle_df['year']==2024) &
3             (vehicle_df['engine']=='24V PDI DOHC Flexible Fuel') &
4             (vehicle_df['fuel']=='E85 Flex Fuel') &
5             (vehicle_df['body']=='Cargo Van') &
6             (vehicle_df['drivetrain']=='Rear-wheel Drive')]
```

Out[39]:

	name	description	make	model	type	year	price	engine	cylinders	fuel	r
31	2024 Ford Transit-250 148 WB Medium Roof Cargo	Load Area Protection Package,Reverse Sensing S...	Ford	Transit-250	New	2024	53595.0	24V PDI DOHC Flexible Fuel	6.0	E85 Flex Fuel	
877	2024 Ford Transit-150 Base	unknown	Ford	Transit-150	New	2024	52640.0	24V PDI DOHC Flexible Fuel	6.0	E85 Flex Fuel	
909	2024 Ford Transit-250 148 WB Medium Roof Cargo	Dealer Discount of \$4,649 off MSRP 2024 Ford T...	Ford	Transit-250	New	2024	50486.0	24V PDI DOHC Flexible Fuel	6.0	E85 Flex Fuel	
940	2024 Ford Transit-350 148 WB Medium Roof Cargo	See what's behind you with the back up camera ...	Ford	Transit-350	New	2024	52530.0	24V PDI DOHC Flexible Fuel	6.0	E85 Flex Fuel	

- Based on above results, for make=Ford, we will replace empty value in field: transmission as Automatic.

```
In [40]: 1 vehicle_df["transmission"].fillna("Automatic", inplace = True)
```

```
In [41]: 1 vehicle_df.isna().sum()
```

```
Out[41]: name                0
description              0
make                    0
model                   0
type                    0
year                    0
price                   0
engine                  0
cylinders               0
fuel                   0
mileage                 0
transmission            0
trim                    0
body                    0
doors                   0
exterior_color          0
interior_color          0
drivetrain              0
dtype: int64
```

```
In [42]: 1 vehicle_df[vehicle_df['trim'].isnull()]
```

```
Out[42]:  name  description  make  model  type  year  price  engine  cylinders  fuel  mileage  transn
```

```
In [43]: 1 vehicle_df.loc[(vehicle_df['make']=='Ford') &
2             (vehicle_df['year']==2023) &
3             (vehicle_df['engine']=='24V GDI DOHC Twin Turbo') &
4             (vehicle_df['fuel']=='Gasoline') &
5             (vehicle_df['body']=='Cargo Van') &
6             (vehicle_df['drivetrain']=='Rear-wheel Drive')]
```

```
Out[43]:  name  description  make  model  type  year  price  engine  cylinders  fuel  mil
```

793	2023 Ford Transit-350 Base	Golden Circle Ford Lincoln has been family own...	Ford	Transit-350	New	2023	57000.0	24V GDI DOHC Twin Turbo	6.0	Gasoline	5
805	2023 Ford Transit-350	Lynn Layton Ford is North Alabama's Fastest Gr...	Ford	Transit-350	New	2023	54525.0	24V GDI DOHC Twin Turbo	6.0	Gasoline	

```
In [44]: 1 vehicle_df.loc[(vehicle_df['make']=='Ford') &
2             (vehicle_df['year']==2023) &
3             (vehicle_df['engine']=='24V GDI DOHC Twin Turbo') &
4             (vehicle_df['fuel']=='Gasoline') &
5             (vehicle_df['body']=='Cargo Van')]
```

```
Out[44]:
```

	name	description	make	model	type	year	price	engine	cylinders	fuel	mil
793	2023 Ford Transit-350 Base	Golden Circle Ford Lincoln has been family own...	Ford	Transit-350	New	2023	57000.0	24V GDI DOHC Twin Turbo	6.0	Gasoline	5
805	2023 Ford Transit-350	Lynn Layton Ford is North Alabama's Fastest Gr...	Ford	Transit-350	New	2023	54525.0	24V GDI DOHC Twin Turbo	6.0	Gasoline	

```
In [45]: 1 vehicle_df.loc[(vehicle_df['make']=='Ford') &
2             (vehicle_df['year']==2023) &
3             (vehicle_df['engine']=='24V GDI DOHC Twin Turbo') &
4             (vehicle_df['fuel']=='Gasoline') ]
```

```
Out[45]:
```

	name	description	make	model	type	year	price	engine	cylinders	fuel	mil
793	2023 Ford Transit-350 Base	Golden Circle Ford Lincoln has been family own...	Ford	Transit-350	New	2023	57000.0	24V GDI DOHC Twin Turbo	6.0	Gasoline	5
805	2023 Ford Transit-350	Lynn Layton Ford is North Alabama's Fastest Gr...	Ford	Transit-350	New	2023	54525.0	24V GDI DOHC Twin Turbo	6.0	Gasoline	

- Based on above records, we replace the empty value for field: trim = Base.

```
In [46]: 1 vehicle_df["trim"].fillna("Base", inplace = True)
```

```
In [47]: 1 vehicle_df.isna().sum()
```

```
Out[47]: name          0
description  0
make        0
model       0
type        0
year        0
price       0
engine      0
cylinders   0
fuel        0
mileage     0
transmission 0
trim        0
body        0
doors       0
exterior_color 0
interior_color 0
drivetrain  0
dtype: int64
```

```
In [48]: 1 vehicle_df[vehicle_df['body'].isnull()]
```

```
Out[48]:
```

	name	description	make	model	type	year	price	engine	cylinders	fuel	mileage	transmission
--	------	-------------	------	-------	------	------	-------	--------	-----------	------	---------	--------------

```
In [49]: 1 vehicle_df.loc[(vehicle_df['make']=='Dodge') &
2                 (vehicle_df['year']==2024) &
3                 (vehicle_df['cylinders']==4.0) &
4                 (vehicle_df['fuel']=='Gasoline') &
5                 (vehicle_df['trim']=='R/T EAWD') &
6                 (vehicle_df['transmission']=='6-Speed Automatic') &
7                 (vehicle_df['drivetrain']=='All-wheel Drive')]
```

```
Out[49]:
```

	name	description	make	model	type	year	price	engine	cylinders	fuel
55	2024 Dodge Hornet R/T EAWD	CALL 239-424-7514 to set up a VIP APPOINTMENT!...	Dodge	Hornet	New	2024	42855.0	gasoline direct injection, DOHC, Multiair va...	4.0	Gasoli
164	2024 Dodge Hornet R/T EAWD	Fresh Arrivall, CALL 239-424-7514 to set up a ...	Dodge	Hornet	New	2024	41497.0	gasoline direct injection, DOHC, Multiair va...	4.0	Gasoli
235	2024 Dodge Hornet R/T EAWD	CALL 239-424-7514 to set up a VIP APPOINTMENT!...	Dodge	Hornet	New	2024	41036.0	gasoline direct injection, DOHC, Multiair va...	4.0	Gasoli

In [50]:

```

1 vehicle_df.loc[(vehicle_df['make']=='Dodge') &
2               (vehicle_df['year']==2024) &
3               (vehicle_df['cylinders']==4.0) &
4               (vehicle_df['fuel']=='Gasoline') &
5               (vehicle_df['trim']=='R/T EAWD') &
6               (vehicle_df['transmission']=='6-Speed Automatic' )]
```

Out[50]:

	name	description	make	model	type	year	price	engine	cylinders	fu
55	2024 Dodge Hornet R/T EAWD	CALL 239-424- 7514 to set up a VIP APPOINTMENT!...	Dodge	Hornet	New	2024	42855.0	4 gasoline direct injection, DOHC, Multiair va...	4.0	Gasoli
164	2024 Dodge Hornet R/T EAWD	Fresh Arrivall, CALL 239-424- 7514 to set up a ...	Dodge	Hornet	New	2024	41497.0	4 gasoline direct injection, DOHC, Multiair va...	4.0	Gasoli
235	2024 Dodge Hornet R/T EAWD	CALL 239-424- 7514 to set up a VIP APPOINTMENT!...	Dodge	Hornet	New	2024	41036.0	4 gasoline direct injection, DOHC, Multiair va...	4.0	Gasoli

In [51]:

```

1 vehicle_df.loc[(vehicle_df['make']=='Dodge') &
2               (vehicle_df['year']==2024) &
3               (vehicle_df['cylinders']==4.0) &
4               (vehicle_df['fuel']=='Gasoline') &
5               (vehicle_df['trim']=='R/T EAWD')]
```

Out[51]:

	name	description	make	model	type	year	price	engine	cylinders	fu
55	2024 Dodge Hornet R/T EAWD	CALL 239-424- 7514 to set up a VIP APPOINTMENT!...	Dodge	Hornet	New	2024	42855.0	4 gasoline direct injection, DOHC, Multiair va...	4.0	Gasoli
164	2024 Dodge Hornet R/T EAWD	Fresh Arrivall, CALL 239-424- 7514 to set up a ...	Dodge	Hornet	New	2024	41497.0	4 gasoline direct injection, DOHC, Multiair va...	4.0	Gasoli
235	2024 Dodge Hornet R/T EAWD	CALL 239-424- 7514 to set up a VIP APPOINTMENT!...	Dodge	Hornet	New	2024	41036.0	4 gasoline direct injection, DOHC, Multiair va...	4.0	Gasoli

```
In [52]: 1 vehicle_df.loc[(vehicle_df['make']=='Dodge') &
2             (vehicle_df['year']==2024) &
3             (vehicle_df['cylinders']==4.0) &
4             (vehicle_df['fuel']=='Gasoline')]
```

Out[52]:

	name	description	make	model	type	year	price	engine	cylin
55	2024 Dodge Hornet R/T EAWD	CALL 239-424-7514 to set up a VIP APPOINTMENT!...	Dodge	Hornet	New	2024	42855.0	4 gasoline direct injection, DOHC, Multiair va...	
56	2024 Dodge Hornet GT	Check out this great value! It comes equipped ...	Dodge	Hornet	New	2024	34990.0	16V GDI DOHC Turbo	
136	2024 Dodge Hornet GT	Lane Keeping Assist,Keyless Start,Rear Spoiler...	Dodge	Hornet	New	2024	38575.0	16V GDI DOHC Turbo	
143	2024 Dodge Hornet GT ΔWN	CALL 239-424-7514 to set up a VIP APPOINTMENT!...	Dodge	Hornet	New	2024	36532.0	ne 2L I-4 gasoline direct injection, DOHC, var...	

- Based on results obtained, we will replace empty value in field: body = SUV.

```
In [53]: 1 vehicle_df.loc[(vehicle_df['make']=='INFINITI') &
2             (vehicle_df['year']==2024) &
3             (vehicle_df['cylinders']==4.0) &
4             (vehicle_df['fuel']=='Gasoline') &
5             (vehicle_df['trim']=='SPORT') &
6             (vehicle_df['transmission']==' (CVT) CONT VAR.') &
7             (vehicle_df['drivetrain']=='Front-wheel Drive')]
```

Out[53]:

	name	description	make	model	type	year	price	engine	cylinders	fuel
687	2024 INFINITI QX50 SPORT	Radiant White 2024 INFINITI QX50 SPORT 12 Spea...	INFINITI	QX50	New	2024	49404.0	ER	4.0	Gasoline

```
In [54]: 1 vehicle_df.loc[(vehicle_df['make']=='INFINITI') &
2               (vehicle_df['year']==2024) &
3               (vehicle_df['cylinders']==4.0) &
4               (vehicle_df['fuel']=='Gasoline') &
5               (vehicle_df['trim']=='SPORT') &
6               (vehicle_df['transmission']=='CVT CONT VAR.')] ]
```

```
Out[54]:
```

	name	description	make	model	type	year	price	engine	cylinders	fuel
		Radiant								
	2024	White 2024								
687	INFINITI	INFINITI	INFINITI	QX50	New	2024	49404.0	ER	4.0	Gasoline
	QX50	QX50								
	SPORT	SPORT 12								
		Spea...								

```
In [55]: 1 vehicle_df.loc[(vehicle_df['make']=='INFINITI') &
2               (vehicle_df['year']==2024) &
3               (vehicle_df['cylinders']==4.0) &
4               (vehicle_df['fuel']=='Gasoline') &
5               (vehicle_df['trim']=='SPORT') ]
```

```
Out[55]:
```

	name	description	make	model	type	year	price	engine	cylinders	fuel
687	2024 INFINITI QX50 SPORT	Radiant White 2024 INFINITI QX50 SPORT 12 Spea...	INFINITI	QX50	New	2024	49404.0	ER	4.0	Gasoline

```
In [56]: 1 vehicle_df.loc[(vehicle_df['make']=='INFINITI') &
2               (vehicle_df['year']==2024) &
3               (vehicle_df['cylinders']==4.0) &
4               (vehicle_df['fuel']=='Gasoline') ]
```

```
Out[56]:
```

	name	description	make	model	type	year	price	engine	cylinders	fuel
167	2024 INFINITI QX50 LUXE	Vaden is pleased to offer 2024 INFINITI QX50 L...	INFINITI	QX50	New	2024	48350.0	o 2L I-4 port/direct injection, DOHC, variable...	4.0	Gasoline
335	2024 INFINITI QX50 LUXE	Vaden is pleased to offer 2024 INFINITI QX50 L...	INFINITI	QX50	New	2024	45055.0	o 2L I-4 port/direct injection, DOHC, variable...	4.0	Gasoline
687	2024 INFINITI QX50 SPORT	Radiant White 2024 INFINITI QX50 SPORT 12 Spea...	INFINITI	QX50	New	2024	49404.0	ER	4.0	Gasoline
799	2024 INFINITI QX50 LUXE	Vaden is pleased to offer 2024 INFINITI QX50 L...	INFINITI	QX50	New	2024	46855.0	o 2L I-4 port/direct injection, DOHC, variable...	4.0	Gasoline

- Based on results obtained, we will replace empty value in field: body = SUV.

```
In [57]: 1 vehicle_df["body"].fillna("SUV", inplace = True)
```

```
In [58]: 1 vehicle_df.isna().sum()
```

```
Out[58]: name          0
description  0
make        0
model       0
type        0
year        0
price       0
engine      0
cylinders   0
fuel        0
mileage     0
transmission 0
trim        0
body        0
doors       0
exterior_color 0
interior_color 0
drivetrain  0
dtype: int64
```

```
In [59]: 1 vehicle_df[vehicle_df['exterior_color'].isnull()]
```

```
Out[59]:  name description make model type year price engine cylinders fuel mileage transn
```

```
In [60]: 1 vehicle_df.loc[(vehicle_df['make']=='Jeep') &
2               (vehicle_df['year']==2024) &
3               (vehicle_df['cylinders']==6.0) &
4               (vehicle_df['fuel']=='Gasoline') &
5               (vehicle_df['body']=='SUV') &
6               (vehicle_df['interior_color']=='Black')]
```

```
Out[60]:
```

	name	description	make	model	type	year	price	engine	cylinders	
6	2024 Jeep Wagoneer Base	Alloy Wheels / Premium Wheels, Apple CarPlay ...	Jeep	Wagoneer	New	2024	63862.0	24V GDI DOHC Twin Turbo	6.0	G
73	2024 Jeep Wagoneer Base	2024 Jeep WagoneerOur Customer Relations Team ...	Jeep	Wagoneer	New	2024	63057.0	24V GDI DOHC Twin Turbo	6.0	G
92	2024 Jeep Wagoneer Base	Alloy Wheels / Premium Wheels, Apple CarPlay ...	Jeep	Wagoneer	New	2024	60555.0	24V GDI DOHC Twin Turbo	6.0	G
	2024 Jeep	Germain of Sidney is						ar 3.6L V-6 DOHC		

```
In [61]: 1 vehicle_df.loc[(vehicle_df['make']=='Jeep') &
2               (vehicle_df['year']==2024) &
3               (vehicle_df['cylinders']==6.0) &
4               (vehicle_df['fuel']=='Gasoline') &
5               (vehicle_df['body']=='SUV') ]
```

Out[61]:

	name	description	make	model	type	year	price	engine	cylinders	1
0	2024 Jeep Wagoneer Series II	Heated Leather Seats, Nav System, Moonroof, Th...	Jeep	Wagoneer	New	2024	74600.0	24V GDI DOHC Twin Turbo	6.0	Gasoline
1	2024 Jeep Grand Cherokee Laredo	Al West is committed to offering every custome...	Jeep	Grand Cherokee	New	2024	50170.0	OHV	6.0	Gasoline
6	2024 Jeep Wagoneer Base	Alloy Wheels / Premium Wheels, Apple CarPlay ...	Jeep	Wagoneer	New	2024	63862.0	24V GDI DOHC Twin Turbo	6.0	Gasoline
9	2024 Jeep Grand Cherokee Altitude	Silver Zynith 2024 Jeep Grand Cherokee Laredo ...	Jeep	Grand Cherokee	New	2024	42773.0	3.6L V-6 DOHC, variable valve control, regul...	6.0	Gasoline
13	2024 Jeep Grand Cherokee Altitude	Check out this 2024! Demonstrating that econom...	Jeep	Grand Cherokee	New	2024	44022.0	3.6L V-6 DOHC, variable valve control, regul...	6.0	Gasoline
...	...	...	...	...	...	...	...	...	...	...
970	2024 Jeep Wagoneer Series II	Heated Leather Seats, NAV, Sunroof, 3rd Row Se...	Jeep	Wagoneer	New	2024	74625.0	24V GDI DOHC Twin Turbo	6.0	Gasoline
973	2024 Jeep Grand Cherokee L Altitude	Price includes Retail Bonus Cash for \$500 Mid...	Jeep	Grand Cherokee L	New	2024	49836.0	3.6L V-6 DOHC, variable valve control, regul...	6.0	Gasoline
983	2024 Jeep Wagoneer L Base	Navigation, 3rd Row Seat, Heated Leather Seats...	Jeep	Wagoneer L	New	2024	66770.0	24V GDI DOHC Twin Turbo	6.0	Gasoline
995	2024 Jeep Grand Cherokee L Limited	Price includes Retail Bonus Cash for \$1 000 M...	Jeep	Grand Cherokee L	New	2024	59674.0	3.6L V-6 DOHC, variable valve control, regul...	6.0	Gasoline
999	2024 Jeep Wagoneer Base	The ALL New Friendship CDJR of Johnson City of...	Jeep	Wagoneer	New	2024	69085.0	24V GDI DOHC Twin Turbo	6.0	Gasoline

89 rows × 18 columns

The values for exterior_color are extremely scattered.

```
In [62]: 1 vehicle_df.loc[(vehicle_df['make']=='Acura') &
2             (vehicle_df['year']==2024) &
3             (vehicle_df['cylinders']==0.0) &
4             (vehicle_df['fuel']=='Electric') &
5             (vehicle_df['body']=='SUV')]
```

Out[62]:

	name	description	make	model	type	year	price	engine	cylinders	fuel	mileage
		2024									
137	Acura ZDX A-SPEC	unknown	Acura	ZDX	New	2024	69850.0	c	0.0	Electric	0

```
In [63]: 1 vehicle_df.loc[(vehicle_df['make']=='Acura') &
2             (vehicle_df['year']==2024) &
3             (vehicle_df['fuel']=='Electric') &
4             (vehicle_df['body']=='SUV')]
```

Out[63]:

	name	description	make	model	type	year	price	engine	cylinders	fuel	mileage
		2024									
137	Acura ZDX A-SPEC	unknown	Acura	ZDX	New	2024	69850.0	c	0.0	Electric	0

```
In [64]: 1 vehicle_df.loc[(vehicle_df['make']=='Acura') &
2               (vehicle_df['year']==2024) &
3               (vehicle_df['body']=='SUV')]
```

Out[64]:

	name	description	make	model	type	year	price	engine	cylinders	fuel
44	2024 Acura MDX Advance	Gray 2024 Acura MDX Advance 10-Speed Automatic...	Acura	MDX	New	2024	65595.0	24V GDI SOHC	6.0	Gasoline
71	2024 Acura MDX Base	Black 2024 Acura MDX 3.5L 10-Speed Automatic V...	Acura	MDX	New	2024	51645.0	24V GDI SOHC	6.0	Gasoline
77	2024 Acura MDX Base	White 2024 Acura MDX 3.5L 10-Speed Automatic V...	Acura	MDX	New	2024	51645.0	24V GDI SOHC	6.0	Gasoline
137	2024 Acura ZDX A-SPEC	unknown	Acura	ZDX	New	2024	69850.0	c	0.0	Electric
161	2024 Acura MDX Technology	White 2024 Acura MDX Technology 10-Speed Autom...	Acura	MDX	New	2024	56155.0	24V GDI SOHC	6.0	Gasoline
245	2024 Acura MDX Base	Gray 2024 Acura MDX 3.5L 10-Speed Automatic V6...	Acura	MDX	New	2024	52950.0	24V GDI SOHC	6.0	Gasoline
286	2024 Acura MDX Technology	Gray 2024 Acura MDX Technology 10-Speed Automa...	Acura	MDX	New	2024	56345.0	24V GDI SOHC	6.0	Gasoline
407	2024 Acura MDX Technology	Gray 2024 Acura MDX Technology 10-Speed Automa...	Acura	MDX	New	2024	57650.0	24V GDI SOHC	6.0	Gasoline
414	2024 Acura MDX Technology	Gray 2024 Acura MDX Technology 10-Speed Automa...	Acura	MDX	New	2024	56800.0	24V GDI SOHC	6.0	Gasoline
478	2024 Acura MDX Technology	Black 2024 Acura MDX Technology 10-Speed Autom...	Acura	MDX	New	2024	56800.0	24V GDI SOHC	6.0	Gasoline
675	2024 Acura MDX Base	Silver 2024 Acura MDX 3.5L 10-Speed Automatic ...	Acura	MDX	New	2024	53700.0	24V GDI SOHC	6.0	Gasoline


```
In [68]: 1 vehicle_df.loc[(vehicle_df['make']=='Mercedes-Benz') &
2              (vehicle_df['year']==2024)]
```

```
Out[68]:
```

	name	description	make	model	type	year	price	engine	cyli
86	2024 Mercedes-Benz EQE 350+ Base	unknown	Mercedes-Benz	EQE 350+	New	2024	76535.0		c
107	2024 Mercedes-Benz Sprinter 2500 Standard Roof	2024 Mercedes-Benz Sprinter 2500 Cargo 144 WB ...	Mercedes-Benz	Sprinter 2500	New	2024	60061.0	16V DDI DOHC Turbo Diesel	
121	2024 Mercedes-Benz EQE 350+ Base	unknown	Mercedes-Benz	EQE 350+	New	2024	79765.0		c

Since we are unable to ascertain values for the field: exterior_color, we will replace all empty records with string unknown.

```
In [69]: 1 vehicle_df["exterior_color"].fillna("unknown", inplace = True)
```

```
In [70]: 1 vehicle_df.isna().sum()
```

```
Out[70]: name          0
description  0
make        0
model       0
type        0
year        0
price       0
engine      0
cylinders   0
fuel        0
mileage     0
transmission 0
trim        0
body        0
doors       0
exterior_color 0
interior_color 0
drivetrain  0
dtype: int64
```

```
In [71]: 1 vehicle_df[vehicle_df['interior_color'].isnull()]
```

```
Out[71]:
```

	name	description	make	model	type	year	price	engine	cylinders	fuel	mileage	transn
--	------	-------------	------	-------	------	------	-------	--------	-----------	------	---------	--------

```
In [72]: 1 vehicle_df["interior_color"].fillna("unknown", inplace = True)
```

```
In [73]: 1 vehicle_df.isna().sum()
```

```
Out[73]: name                0
description                0
make                      0
model                    0
type                     0
year                     0
price                    0
engine                   0
cylinders                0
fuel                     0
mileage                  0
transmission             0
trim                     0
body                     0
doors                   0
exterior_color           0
interior_color           0
drivetrain               0
dtype: int64
```

Thus, there are no null values in any fields.

```
In [74]: 1 vehicle_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1002 entries, 0 to 1001
Data columns (total 18 columns):
#   Column                Non-Null Count  Dtype
---  -
0   name                  1002 non-null   object
1   description            1002 non-null   object
2   make                  1002 non-null   object
3   model                 1002 non-null   object
4   type                  1002 non-null   object
5   year                  1002 non-null   int64
6   price                 1002 non-null   float64
7   engine                1002 non-null   object
8   cylinders              1002 non-null   float64
9   fuel                  1002 non-null   object
10  mileage                1002 non-null   float64
11  transmission           1002 non-null   object
12  trim                   1002 non-null   object
13  body                   1002 non-null   object
14  doors                  1002 non-null   float64
15  exterior_color         1002 non-null   object
16  interior_color         1002 non-null   object
17  drivetrain             1002 non-null   object
dtypes: float64(4), int64(1), object(13)
memory usage: 141.0+ KB
```

In all categorical fields, we trim leading and trailing white spaces.

```
In [75]: 1 vehicle_df["name"] = vehicle_df["name"].str.strip()
2 vehicle_df["description"] = vehicle_df["description"].str.strip()
3 vehicle_df["make"] = vehicle_df["make"].str.strip()
4 vehicle_df["model"] = vehicle_df["model"].str.strip()
5 vehicle_df["type"] = vehicle_df["type"].str.strip()
6 vehicle_df["engine"] = vehicle_df["engine"].str.strip()
7 vehicle_df["fuel"] = vehicle_df["fuel"].str.strip()
8 vehicle_df["transmission"] = vehicle_df["transmission"].str.strip()
9 vehicle_df["trim"] = vehicle_df["trim"].str.strip()
10 vehicle_df["body"] = vehicle_df["body"].str.strip()
11 vehicle_df["exterior_color"] = vehicle_df["exterior_color"].str.strip()
12 vehicle_df["interior_color"] = vehicle_df["interior_color"].str.strip()
13 vehicle_df["drivetrain"] = vehicle_df["drivetrain"].str.strip()
```

```
In [76]: 1 vehicle_df.head()
```

```
Out[76]:
```

	name	description	make	model	type	year	price	engine	cylinders	fuel
0	2024 Jeep Wagoneer Series II	Heated Leather Seats, Nav System, Moonroof, Th...	Jeep	Wagoneer	New	2024	74600.0	24V GDI DOHC Twin Turbo	6.0	Gasoline
1	2024 Jeep Grand Cherokee Laredo	AI West is committed to offering every custome...	Jeep	Grand Cherokee	New	2024	50170.0	OHV	6.0	Gasoline
2	2024 GMC Yukon XL Denali	unknown	GMC	Yukon XL	New	2024	96410.0	6.2L V-8 gasoline direct injection, variable v...	8.0	Gasoline
3	2023 Dodge Durango Pursuit	White Knuckle Clearcoat 2023 Dodge Durango Pur...	Dodge	Durango	New	2023	46835.0	16V MPFI OHV	8.0	Gasoline
4	2024 RAM 3500 Laramie	2024 Ram 3500 Laramie Billet Silver Metallic C...	RAM	3500	New	2024	81663.0	24V DDI OHV Turbo Diesel	6.0	Diese

In [77]:

1 vehicle_df.tail()

Out[77]:

	name	description	make	model	type	year	price	engine	cylinders
997	2024 Mercedes-Benz Sprinter 2500 Standard Roof	2024 Mercedes-Benz Sprinter 2500 Cargo 144 WB ...	Mercedes-Benz	Sprinter 2500	New	2024	59037.0	16V DDI DOHC Turbo Diesel	4.0
998	2024 Dodge Hornet R/T Plus Eawd	Dealer Comments +++ Price Ends 5/31/2024 +++ A...	Dodge	Hornet	New	2024	49720.0	4 gasoline direct injection, DOHC, Multiair va...	4.0 (
999	2024 Jeep Wagoneer Base	The ALL New Friendship CDJR of Johnson City of...	Jeep	Wagoneer	New	2024	69085.0	24V GDI DOHC Twin Turbo	6.0 (
1000	2024 Nissan Murano SV Intelligent AWD	CVT with Xtronic, AWD.At Todd Nissan Of Torrin...	Nissan	Murano	New	2024	43495.0	6 DOHC, variable valve control, regular unlead...	6.0 (
1001	2024 Chevrolet Silverado 2500 WT	01u 2024 Chevrolet Silverado 2500HD Work Truck...	Chevrolet	Silverado 2500	New	2024	48995.0	8 gasoline direct injection, variable valve co...	8.0 (

Checking for duplicate rows in the dataset

In [78]:

1 print(vehicle_df.duplicated().sum())

24

Removing these 24 duplicate rows from the dataset

In [79]:

```
1 print(vehicle_df.duplicated().sum())
2 vehicle_df.drop_duplicates(inplace=True)
3
```

In [80]:

1 vehicle_df.shape

Out[80]: (978, 18)

```
In [81]: 1 vehicle_df.head()
```

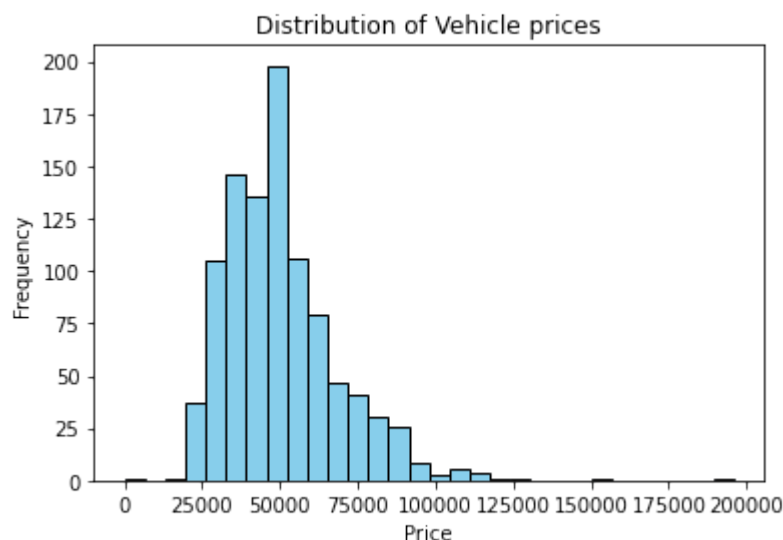
Out[81]:

	name	description	make	model	type	year	price	engine	cylinders	fuel
0	2024 Jeep Wagoneer Series II	Heated Leather Seats, Nav System, Moonroof, Th...	Jeep	Wagoneer	New	2024	74600.0	24V GDI DOHC Twin Turbo	6.0	Gasoline
1	2024 Jeep Grand Cherokee Laredo	Al West is committed to offering every custome...	Jeep	Grand Cherokee	New	2024	50170.0	OHV	6.0	Gasoline
2	2024 GMC Yukon XL Denali	unknown	GMC	Yukon XL	New	2024	96410.0	6.2L V-8 gasoline direct injection, variable v...	8.0	Gasoline
3	2023 Dodge Durango Pursuit	White Knuckle Clearcoat 2023 Dodge Durango Pur...	Dodge	Durango	New	2023	46835.0	16V MPFI OHV	8.0	Gasoline
4	2024 RAM 3500 Laramie	2024 Ram 3500 Laramie Billet Silver Metallic C...	RAM	3500	New	2024	81663.0	24V DDI OHV Turbo Diesel	6.0	Diese

Exploratory Data Analysis

```
In [82]: 1 plt.hist(vehicle_df['price'], bins=30, color='skyblue', edgecolor='black')
2 plt.xlabel('Price')
3 plt.ylabel('Frequency')
4 plt.title('Distribution of Vehicle prices')
```

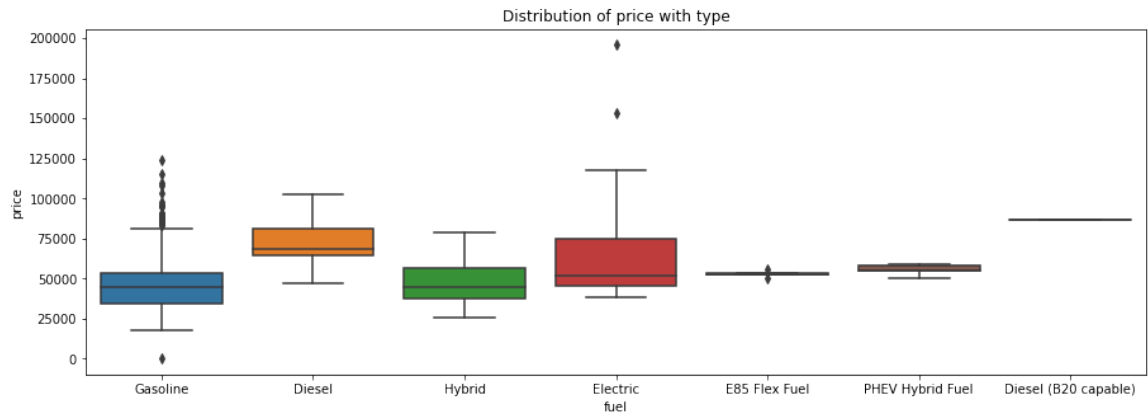
Out[82]: Text(0.5, 1.0, 'Distribution of Vehicle prices')



- We observed positively skewed histogram by taking prices of all vehicles in the dataset.
- Thus, the prices of vehicles are concentrated on lower end of price range: 25000 to 75000.

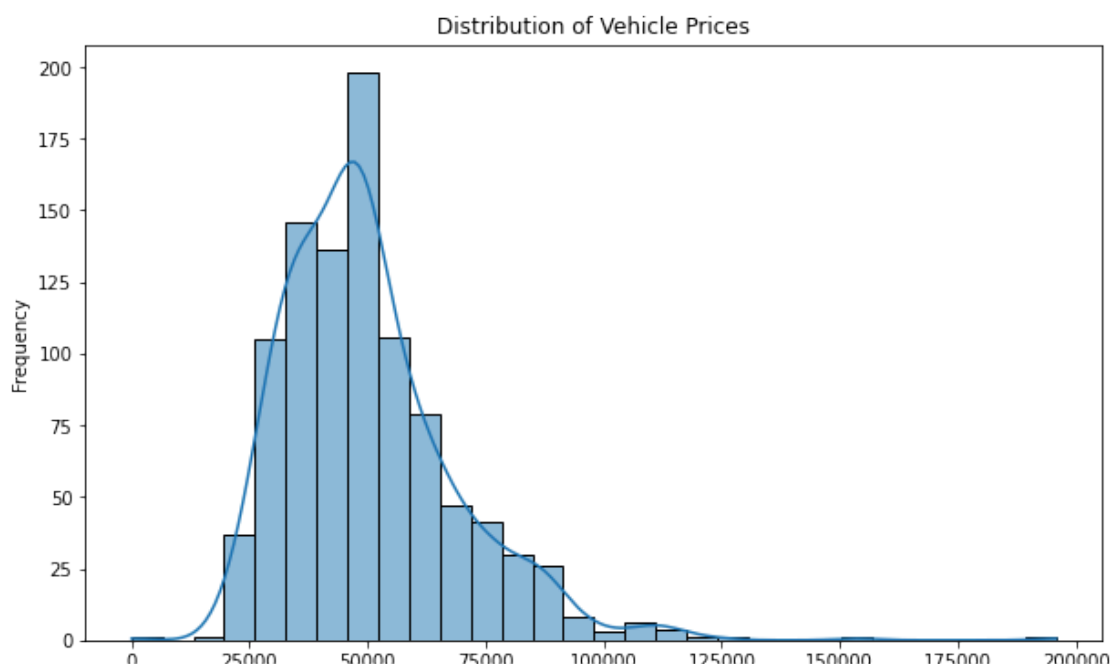
```
In [83]: 1 plt.figure(figsize=(15,5))
2         sns.boxplot(x='fuel',y='price',data=vehicle_df)
3         plt.title('Distribution of price with type')
```

Out[83]: Text(0.5, 1.0, 'Distribution of price with type')



In [84]:

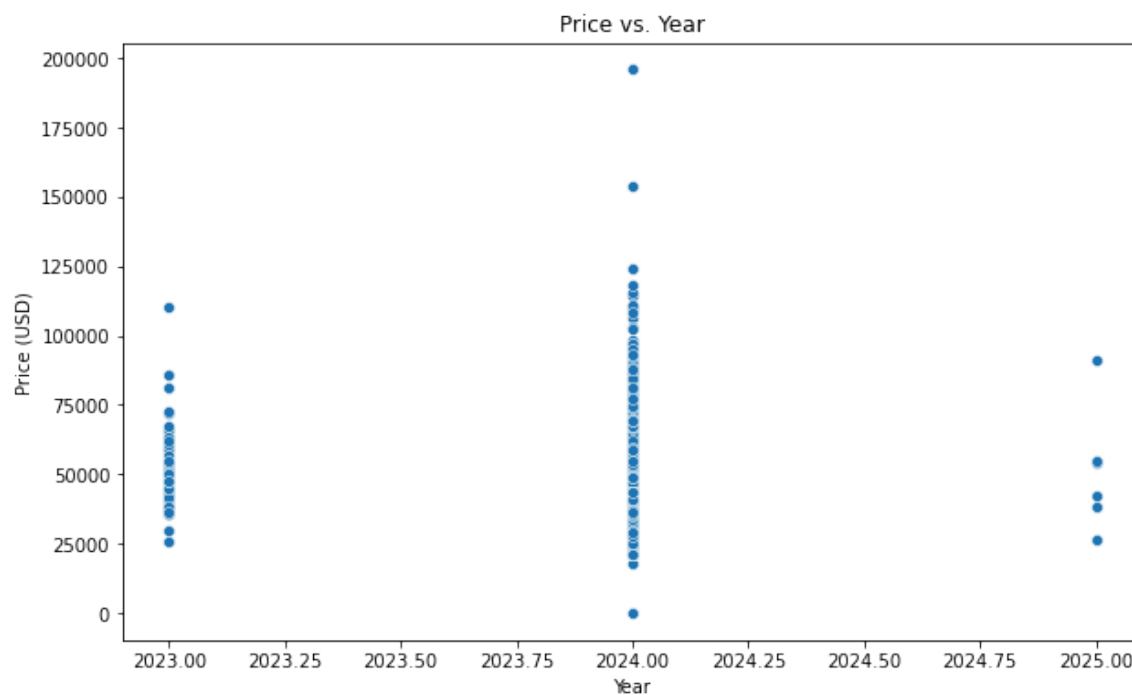
```
1 # Distribution of vehicle prices
2 plt.figure(figsize=(10, 6))
3 sns.histplot(vehicle_df['price'], bins=30, kde=True)
4 plt.title('Distribution of Vehicle Prices')
5 plt.xlabel('Price (USD)')
6 plt.ylabel('Frequency')
7 plt.show()
8
9 # Count of vehicles by make
10 plt.figure(figsize=(12, 8))
11 sns.countplot(data=vehicle_df, y='make', order=vehicle_df['make'].value_counts().index)
12 plt.title('Count of Vehicles by Make')
13 plt.xlabel('Count')
14 plt.ylabel('Make')
15 plt.show()
16
17 # Average price by make
18 plt.figure(figsize=(12, 8))
19 avg_price_by_make = vehicle_df.groupby('make')['price'].mean().sort_values(ascending=False)
20 sns.barplot(x=avg_price_by_make, y=avg_price_by_make.index)
21 plt.title('Average Price by Make')
22 plt.xlabel('Average Price (USD)')
23 plt.ylabel('Make')
24 plt.show()
25
26 # Price distribution by year
27 plt.figure(figsize=(10, 6))
28 sns.boxplot(data=vehicle_df, x='year', y='price')
29 plt.title('Price Distribution by Year')
30 plt.xlabel('Year')
31 plt.ylabel('Price (USD)')
32 plt.show()
33
34 # Distribution of number of cylinders
35 plt.figure(figsize=(10, 6))
36 sns.countplot(data=vehicle_df, x='cylinders')
37 plt.title('Distribution of Number of Cylinders')
38 plt.xlabel('Cylinders')
39 plt.ylabel('Count')
40 plt.show()
```

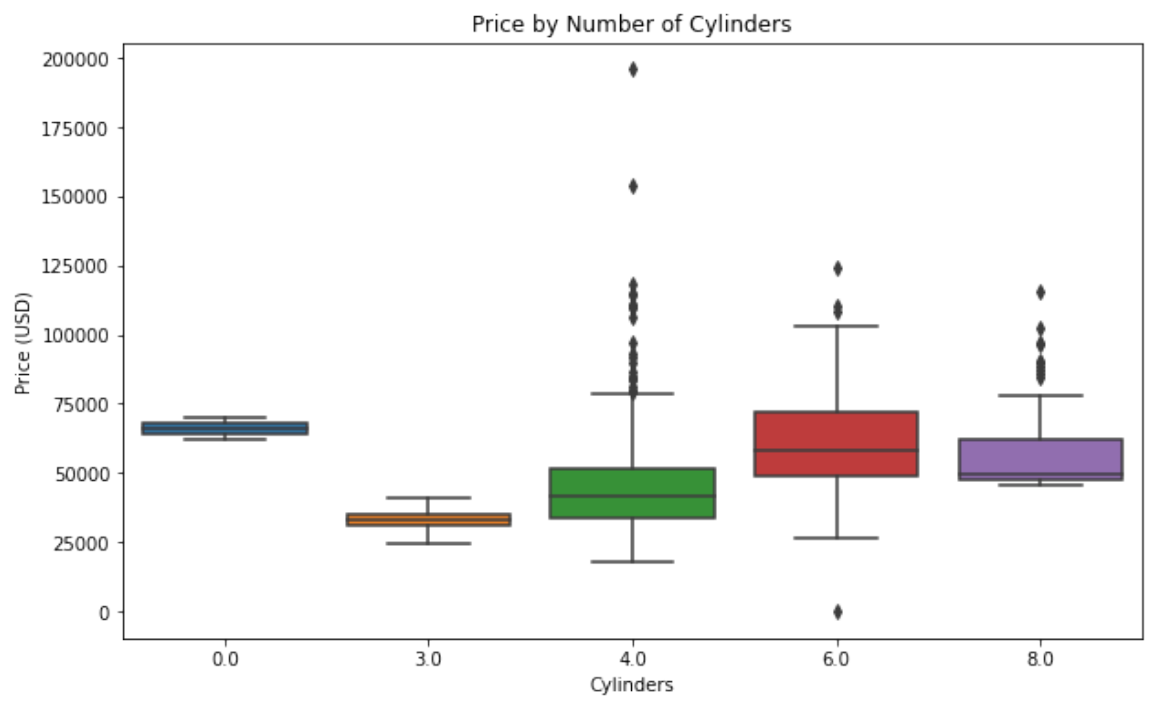


In [97]:

```
1 # Correlation between price and numerical features
2 corr_matrix = vehicle_df.corr()
3 print(corr_matrix['price'].sort_values(ascending=False))
4
5 # Scatter plot of price vs. year
6 plt.figure(figsize=(10, 6))
7 sns.scatterplot(data=vehicle_df, x='year', y='price')
8 plt.title('Price vs. Year')
9 plt.xlabel('Year')
10 plt.ylabel('Price (USD)')
11 plt.show()
12
13 # Boxplot of price by number of cylinders
14 plt.figure(figsize=(10, 6))
15 sns.boxplot(data=vehicle_df, x='cylinders', y='price')
16 plt.title('Price by Number of Cylinders')
17 plt.xlabel('Cylinders')
18 plt.ylabel('Price (USD)')
19 plt.show()
```

```
price      1.000000
cylinders   0.359446
mileage     0.076792
year        0.004668
doors      -0.069131
Name: price, dtype: float64
```

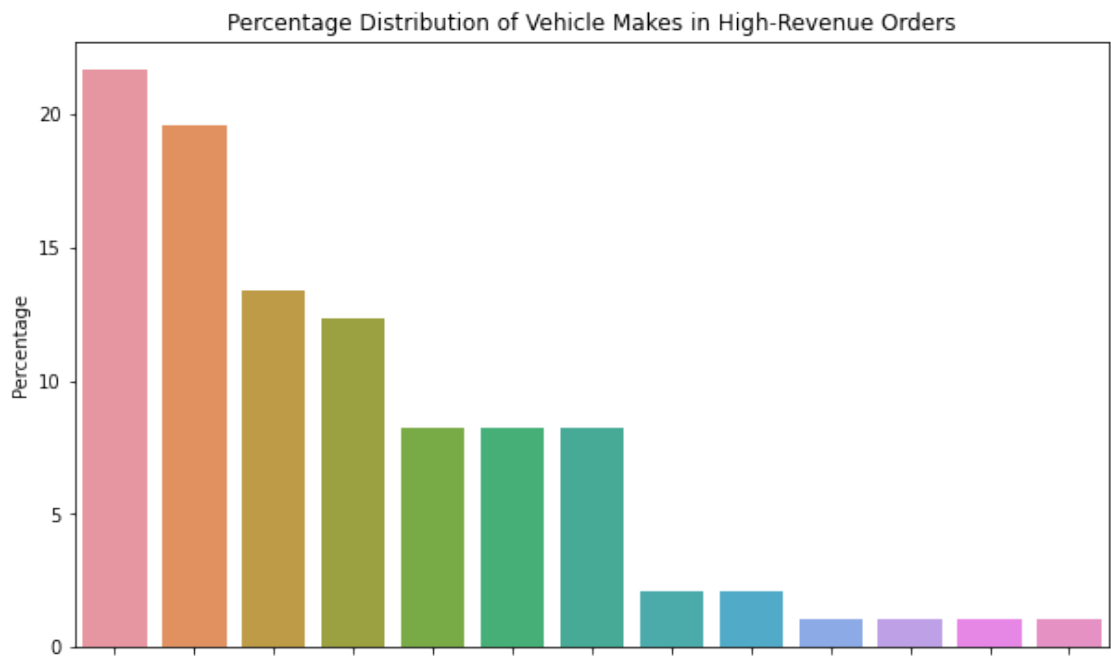




**To check factors contributing to high revenue orders,
we separate the orders above 75th percentile price and
analyse the trend**

In [92]:

```
1 # Identify high-revenue orders (assuming high-revenue is defined as price > 75000)
2 import copy
3 df = copy.deepcopy(vehicle_df)
4
5 high_revenue_threshold = 75000
6 high_revenue_orders = df[df['price'] > high_revenue_threshold]
7
8 # Calculate percentages for categorical features
9 def plot_percentage_distribution(df, column, title):
10     counts = df[column].value_counts()
11     percentages = (counts / counts.sum()) * 100
12     percentages_df = percentages.reset_index()
13     percentages_df.columns = [column, 'percentage']
14
15     plt.figure(figsize=(10, 6))
16     sns.barplot(x=column, y='percentage', data=percentages_df, order=percentages_df[column].sort_values(ascending=False))
17     plt.title(title)
18     plt.xlabel(column)
19     plt.ylabel('Percentage')
20     plt.xticks(rotation=90)
21     plt.show()
22
23 # Factors contributing to high-revenue orders - Make
24 plot_percentage_distribution(high_revenue_orders, 'make', 'Percentage Distribution of Make')
25
26 # Factors contributing to high-revenue orders - Year
27 plot_percentage_distribution(high_revenue_orders, 'year', 'Percentage Distribution of Year')
28
29 # Factors contributing to high-revenue orders - Engine
30 plot_percentage_distribution(high_revenue_orders, 'engine', 'Percentage Distribution of Engine')
31
32 # Factors contributing to high-revenue orders - Cylinders
33 plot_percentage_distribution(high_revenue_orders, 'cylinders', 'Percentage Distribution of Cylinders')
34
35 # Factors contributing to high-revenue orders - type
36 plot_percentage_distribution(high_revenue_orders, 'type', 'Percentage Distribution of Type')
37
38 # Factors contributing to high-revenue orders - model
39 plot_percentage_distribution(high_revenue_orders, 'model', 'Percentage Distribution of Model')
40
41 # Factors contributing to high-revenue orders - fuel
42 plot_percentage_distribution(high_revenue_orders, 'fuel', 'Percentage Distribution of Fuel')
43
44 # Factors contributing to high-revenue orders - transmission
45 plot_percentage_distribution(high_revenue_orders, 'transmission', 'Percentage Distribution of Transmission')
46
47 # Factors contributing to high-revenue orders - doors
48 plot_percentage_distribution(high_revenue_orders, 'doors', 'Percentage Distribution of Doors')
49
50 # Factors contributing to high-revenue orders - exterior_color
51 plot_percentage_distribution(high_revenue_orders, 'exterior_color', 'Percentage Distribution of Exterior Color')
52
53 # Factors contributing to high-revenue orders - trim
54 plot_percentage_distribution(high_revenue_orders, 'trim', 'Percentage Distribution of Trim')
55
```



In [93]:

```
1 corr_matrix = high_revenue_orders.corr()  
2 print(corr_matrix['price'].sort_values(ascending=False))
```

```
price      1.000000  
mileage    0.035655  
doors      0.031993  
year       -0.020179  
cylinders  -0.209875  
Name: price, dtype: float64
```